6 (a) greater binding energy gives rise to release of energy M1
so must be yttrium .... A1 [2]

(b) probability of decay M1
of a nucleus per unit time .... A1 [2]

(c) (i)1 $A = \lambda N$ C1
$3.7 \times 10^6 \times 365 \times 24 \times 3600 = 0.025N$ C1
$N = 4.67 \times 10^{15}$ .... A1 [3]

(i)2 mass = $0.09 \times (4.67 \times 10^{15})/(6.02 \times 10^{23})$ C1
$= 6.98 \times 10^{-10}$ kg .... A1 [2]

(ii) $A = A_0 e^{-\lambda t}$
$A/A_0 = e^{-0.025t}$ C1
$= 0.88$ .... A1 [2]